Jahangirnagar University

Institute of Information Technology

Savar, Dhaka-1342

LAB REPORT - 01

Connecting End Devices (PCs) Using Hub and Switch

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

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Submitted By

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ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

7 May 2026

Lab -1: Connecting End Devices (PCs) Using Hub and Switch

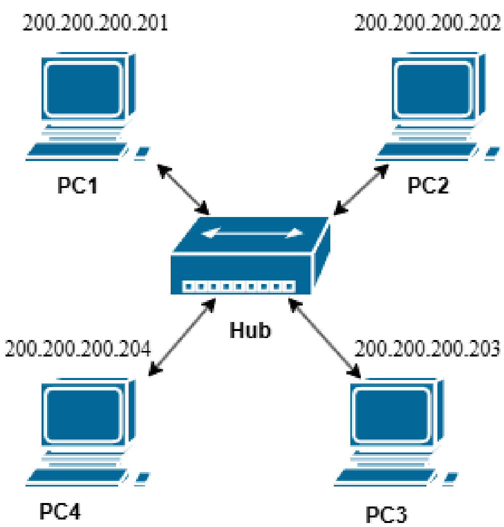
Objective:

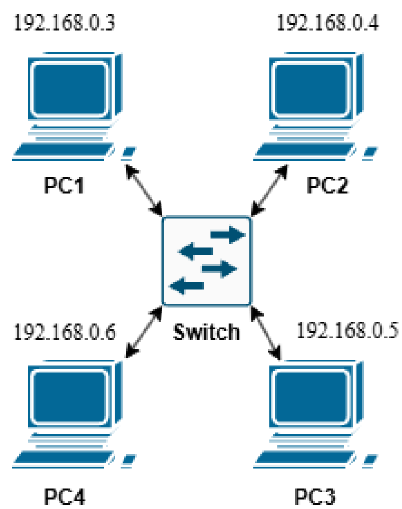
To understand how end devices (PCs) communicate in a network using a hub and a switch, and to observe the differences in data transmission behavior between a hub-based and a switch-based network.

Apparatus:

1. Cisco Packet Tracer
2. 1 Hub
3. 1 Switch
4. 8 End Devices (PCs or Laptops)
5. Copper Straight-Through Ethernet Cable

Network diagram:





Procedure:

1. Open Cisco Packet Tracer and place 1 hub, 1 switch, and 8 PCs on the workspace.

2. Connect PC1 through PC4 to the hub using copper straight-through ethernet cables. Connect PC5 through PC8 to the switch in the same way.

3. Assign static IP addresses to all PCs as follows:

Hub-connected PCs:

Device	IP Address
PC1	200.200.200.201
PC2	200.200.200.202
PC3	200.200.200.203
PC4	200.200.200.204

Switch-connected PCs:

Device	IP Address
PC1	192.168.0.3
PC2	192.168.0.4
PC3	192.168.0.5
PC4	192.168.0.6

Since all PCs within each group belong to the same network, no default gateway is required.

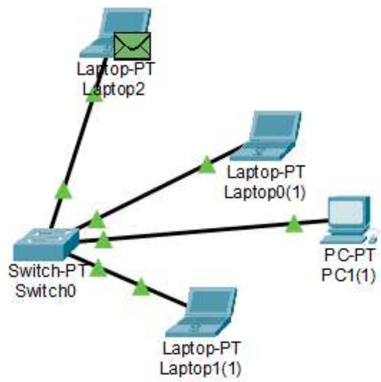
4. To assign an IP address, click on a PC, go to Desktop → IP Configuration, select Static, and enter the IP address and subnet mask (255.255.255.0).

5. Test connectivity by opening the Command Prompt on any PC and running the ping command toward another PC in the same group. For example:

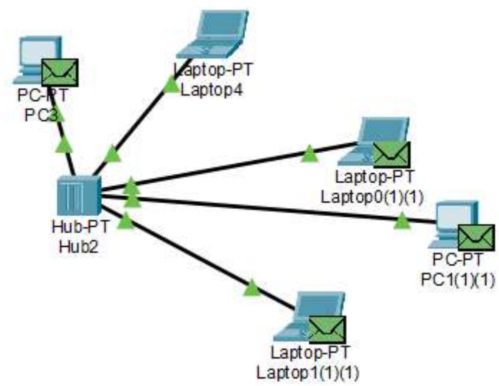
ping 200.200.200.202

6. Observe the behavior: on the hub network, all PCs receive the data packet (broadcast behavior), while on the switch network, data is sent only to the specific destination PC (unicast behavior).

Output:



Laptop-PT
Laptop2
Laptop-PT
Laptop0(1)
PC-PT
PC1(1)
Switch-PT
Switch0
Laptop-PT
Laptop1(1)



PC-PT
PC3
Laptop-PT
Laptop4
Laptop-PT
Laptop0(1)(1)
Hub-PT
Hub2
PC-PT
PC1(1)(1)
Laptop-PT
Laptop1(1)(1)

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LAB REPORT - 03

Connecting End Devices (PCs) Using a Router

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

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Submitted By

Md. Shaon Khan

ID: 1984   Batch: 52   Session: 2022–2023

Date of Submission

7 May 2026

Lab -3: Connecting End Devices (PCs) Using a Router

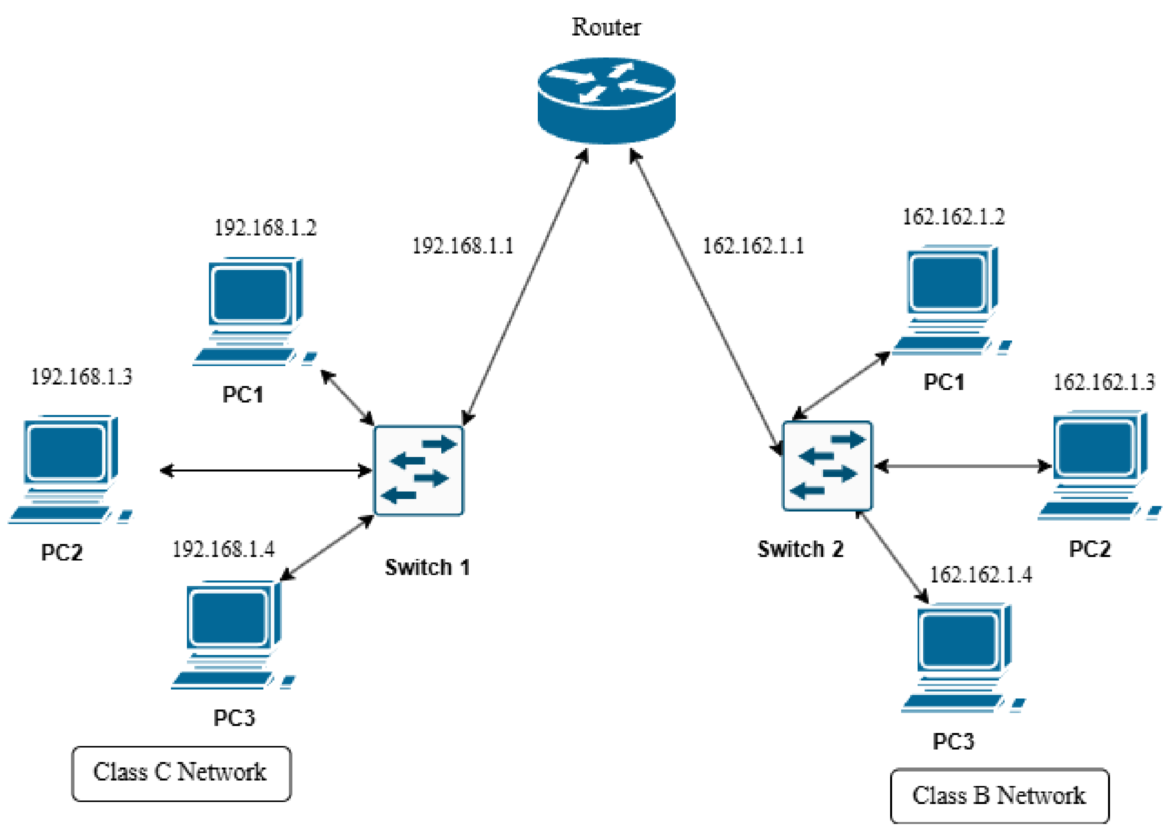
Objective:

To study the role of a router in connecting different networks by configuring end devices (PCs) on separate networks, and to observe how data is transmitted and routed between these networks through proper IP configuration and connectivity testing.

Apparatus:

1. Cisco Packet Tracer
2. Switches
3. 6 End Devices (PCs or Laptops)
4. 1 Router
5. Copper Straight-Through Ethernet Cable

Network diagram:



Procedure:

1. Open Cisco Packet Tracer and place 1 router, 2 switches, and 6 PCs on the workspace.

2. Connect PC1, PC2, and PC3 to Switch1 using copper straight-through ethernet cables. Connect the remaining PC4, PC5, and PC6 to Switch2 in the same way. Then connect Switch1 to one interface of the router (Fa0/0) and Switch2 to the other interface (Fa0/1).

3. Assign static IP addresses to all PCs in their respective networks as follows:

   Switch1 Network (192.168.1.0/24):

Device	IP Address	Default Gateway
PC1	192.168.1.2	192.168.1.1
PC2	192.168.1.3	192.168.1.1
PC3	192.168.1.4	192.168.1.1

   Switch2 Network (162.162.1.0/24):

Device	IP Address	Default Gateway
PC4	162.162.1.2	162.162.1.1
PC5	162.162.1.3	162.162.1.1
PC6	162.162.1.4	162.162.1.1

   Since the two groups of PCs belong to different networks, a default gateway is required for inter-network communication. The default gateway for each PC is the IP address of the router interface connected to its network.

4. Configure the router interfaces via the CLI. Click on the router, open the **CLI** tab, and enter the following commands:

```
Router> enable
Router# configure terminal

Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface FastEthernet0/1
Router(config-if)# ip address 162.162.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end
```

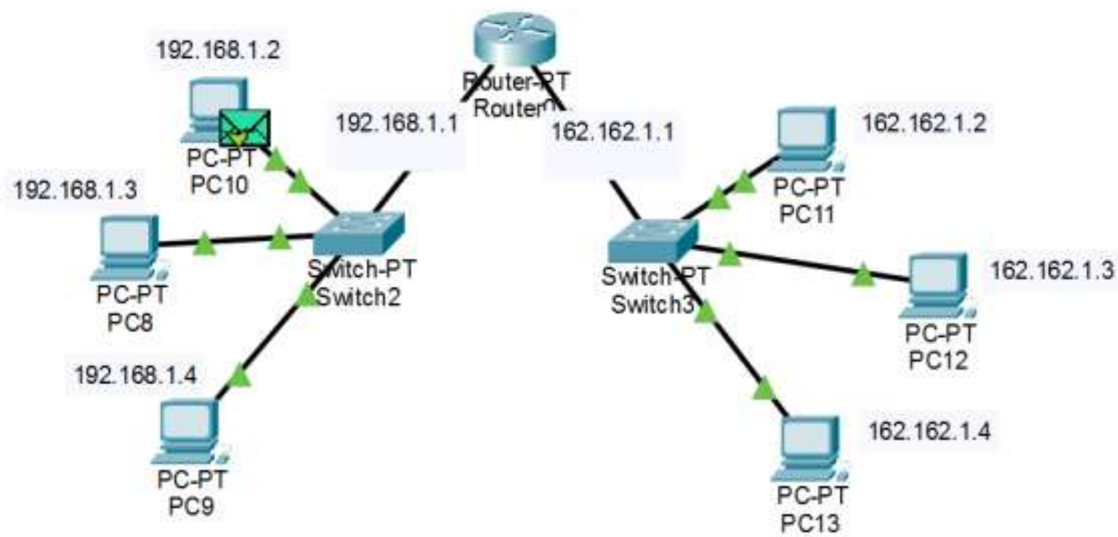
5. To assign IP addresses to the PCs, click on each PC, go to Desktop → IP Configuration, select Static, and enter the IP address, subnet mask (255.255.255.0), and default gateway as listed in the table above.

6. Verify connectivity by opening the Command Prompt on a PC in Switch1's network and pinging a PC in Switch2's network. For example:

   ping 162.162.1.2

7. Observe the ping replies. Successful replies confirm that the router is correctly forwarding packets between the two different networks.

Output:



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LAB REPORT - 04

IP Address Assign Using DHCP

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

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ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

7 May 2026

Lab -4: IP Address Assign Using DHCP

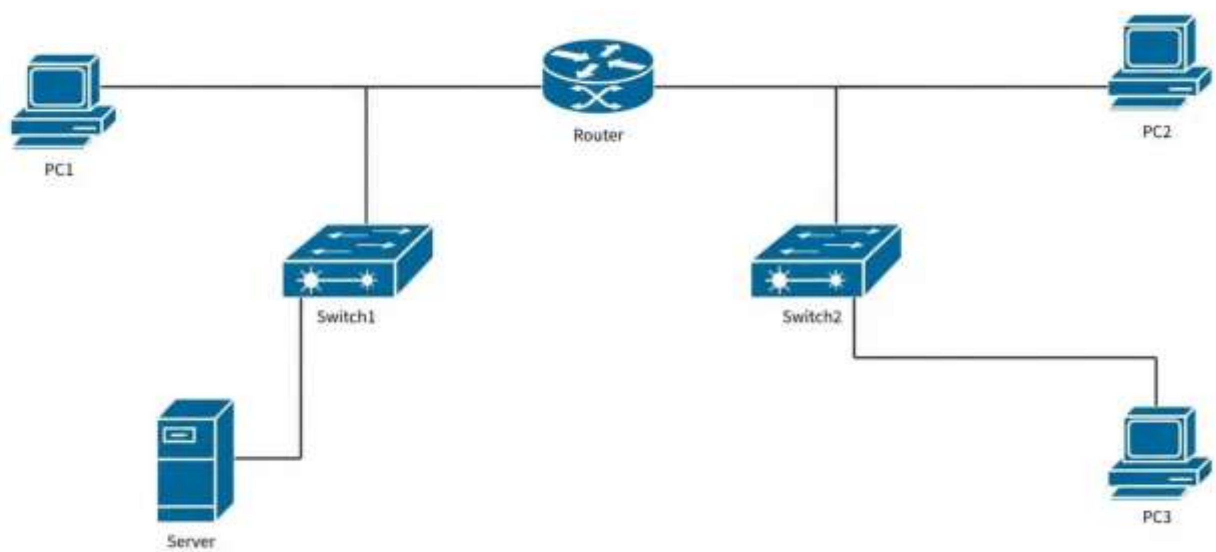
Objective:

To understand how IP addresses are dynamically assigned to end devices using DHCP (Dynamic Host Configuration Protocol), and to observe the automatic IP configuration process in a simulated network environment.

Apparatus:

1. Cisco Packet Tracer
2. Router
3. Switches
4. Server (DHCP Server)
5. end devices (PCs)
6. Copper straight through ethernet wire

Network diagram:



Procedure:

1. Open Cisco Packet Tracer and build the network topology as shown in the network diagram. Place one Router (Router2), two Switches (Switch4 and Switch5), one Server (Server4), and five PCs (PC8, PC9, PC10, and two more) on the workspace.

2. Connect all devices using copper straight-through ethernet cables as follows: PC8 (Fa0) to Switch4 (Fa1/1), Server4 (Fa0) to Switch4 (Fa2/1), Switch4 (Fa0/1) to Router2 (Fa0/0), Router2 (Fa1/0) to Switch5 (Fa0/1), PC9 (Fa0) to Switch5 (Fa1/1), PC10 (Fa0) to Switch5 (Fa2/1).

3. Assign a static IP address to Server4. Click on Server4, go to Desktop > IP Configuration, and enter the following:
   a. IP Address : 192.168.1.100
   b. Subnet Mask : 255.255.255.0
   c. Default GW : 192.168.1.1

4. Configure the router interfaces with static IP addresses. Click on Router2, open the CLI tab, and enter the following commands:
   a. Router> enable
   b. Router# configure terminal
   c. Router(config)# interface FastEthernet0/0
   d. Router(config-if)# ip address 192.168.1.1 255.255.255.0
   e. Router(config-if)# ip helper-address 192.168.1.100
   f. Router(config-if)# no shutdown
   g. Router(config-if)# exit
   h. Router(config)# interface FastEthernet1/0
   i. Router(config-if)# ip address 192.168.2.1 255.255.255.0
   j. Router(config-if)# ip helper-address 192.168.1.100
   k. Router(config-if)# no shutdown
   l. Router(config-if)# end

5. Configure the DHCP service on Server4. Click on Server4, go to Services > DHCP, and set up two address pools as follows:
   a. Pool Name : Network1
   b. Start IP : 192.168.1.2
   c. Subnet Mask : 255.255.255.0
   d. Default GW : 192.168.1.1

   e. Pool Name : Network2
   f. Start IP : 192.168.2.2
   g. Subnet Mask : 255.255.255.0

h. Default GW : 192.168.2.1

6. Enable DHCP service on the server by turning the service ON in the DHCP settings page. Click Add after entering each pool configuration.

7. Set each PC's IP configuration to DHCP mode. Click on each PC, go to Desktop > IP Configuration, and select DHCP. Each PC will automatically receive an IP address, subnet mask, and default gateway from the server. Verify that the assigned addresses match the expected ranges from the diagram (PC8: 192.168.1.2, PC9: 192.168.2.2, PC10: 192.168.2.3).

8. Verify connectivity by opening a command prompt on any PC (Desktop > Command Prompt) and using the ping command to test communication between PCs on both networks. For example:
   a. C:\> ping 192.168.2.2
   b. C:\> ping 192.168.1.2

9. Observe the successful ping replies confirming that DHCP has assigned correct IP addresses and the router is properly forwarding packets between the two networks through the centralized DHCP server.

Output: